Results

Descriptives

Descriptives

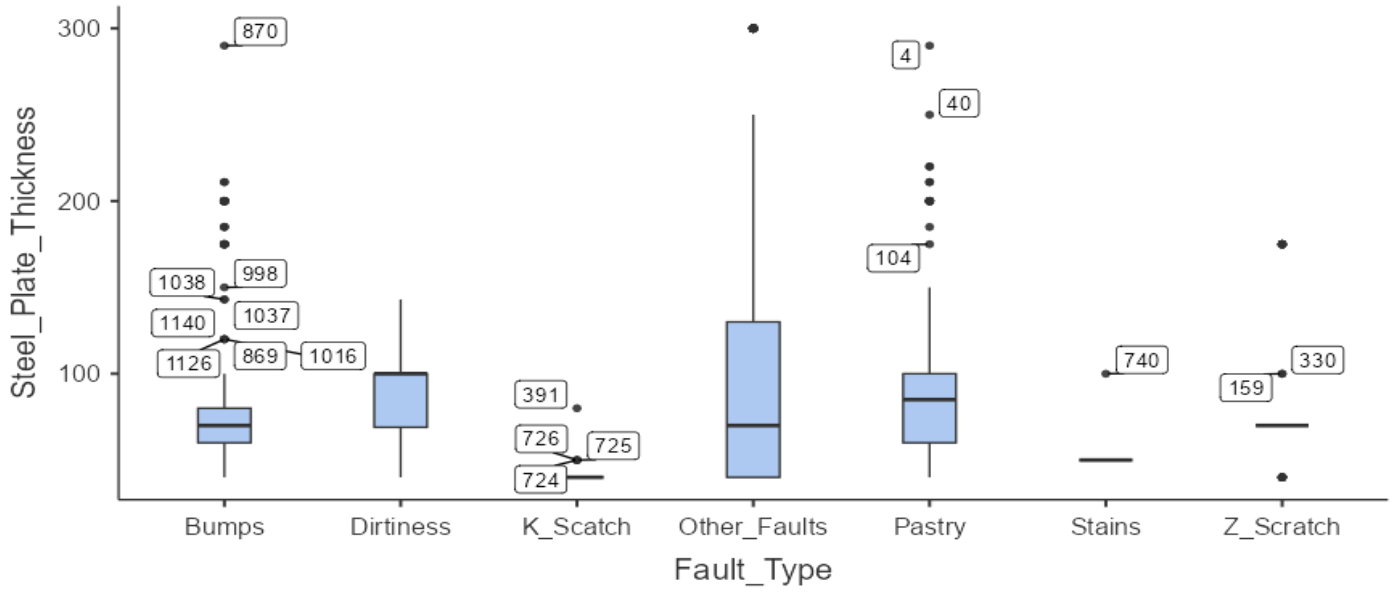
	Fault_Type	Steel_Plate_Thickness
N	Bumps	402
	Dirtiness	55
	K_Scatch	391
	Other_Faults	673
	Pastry	158
	Stains	72
	Z_Scratch	190
Missing	Bumps	0
	Dirtiness	0
	K_Scatch	0
	Other_Faults	0
	Pastry	0
	Stains	0
	Z_Scratch	0
Mean	Bumps	76.9
	Dirtiness	84.5
	K_Scatch	40.2
	Other_Faults	103
	Pastry	92.9
	Stains	50.7
	Z_Scratch	74.6
Median	Bumps	70.0
	Dirtiness	100
	K_Scatch	40
	Other_Faults	70
	Pastry	85.0
	Stains	50.0
	Z_Scratch	70.0
Standard deviation	Bumps	38.0
	Dirtiness	28.7
	K_Scatch	2.20
	Other_Faults	73.9
	Pastry	51.6
	Stains	5.89
	Z_Scratch	24.7
Minimum	Bumps	40
	Dirtiness	40
	K_Scatch	40
	Other_Faults	40
	Pastry	40
	Stains	50
	Z_Scratch	40

Descriptives

Maximum	Bumps	290
	Dirtiness	143
	K_Scatch	80
	Other_Faults	300
	Pastry	290
	Stains	100
	Z_Scratch	175

Plots

Steel_Plate_Thickness



One-Way ANOVA

One-Way ANOVA (Welch's)

	F	df1	df2	p
Steel_Plate_Thickness	283	6	337	<.001

Post Hoc Tests

Games-Howell Post-Hoc Test – Steel_Plate_Thickness

		Bumps	Dirtiness	K_Scatch	Other_Faults	Pastry	Stains	Z_Scratch
Bumps	Mean difference	—	-7.52	36.8	-25.7	-15.97	26.2	2.35
	p-value	—	.586	<.001	<.001	.009	<.001	.972
Dirtiness	Mean difference		—	44.3	-18.1	-8.45	33.8	9.88
	p-value		—	<.001	.004	.745	<.001	.248
K_Scatch	Mean difference			—	-62.4	-52.73	-10.5	-34.40
	p-value			—	<.001	<.001	<.001	<.001
Other_Faults	Mean difference				—	9.70	51.9	28.02
	p-value				—	.455	<.001	<.001
Pastry	Mean difference					—	42.2	18.33
	p-value					—	<.001	.001
Stains	Mean difference						—	-23.88
	p-value						—	<.001
Z_Scratch	Mean difference							—
	p-value							—

Contingency Tables

Contingency Tables

		TypeOfSteel_A300		
Fault_Type		0	1	Total
Bumps	Observed	123	279	402
	% within row	30.6%	69.4%	100.0%
Dirtiness	Observed	46	9	55
	% within row	83.6%	16.4%	100.0%
K_Scatch	Observed	390	1	391
	% within row	99.7%	0.3%	100.0%
Other_Faults	Observed	407	266	673
	% within row	60.5%	39.5%	100.0%
Pastry	Observed	109	49	158
	% within row	69.0%	31.0%	100.0%
Stains	Observed	71	1	72
	% within row	98.6%	1.4%	100.0%
Z_Scratch	Observed	18	172	190
	% within row	9.5%	90.5%	100.0%
Total	Observed	1164	777	1941
	% within row	60.0%	40.0%	100.0%

χ^2 Tests

	Value	df	p
χ^2	667	6	<.001
N	1941		

Descriptives

Descriptives

	Fault_Type	Length_of_Conveyer
N	Bumps	402
	Dirtiness	55
	K_Scatch	391
	Other_Faults	673
	Pastry	158
	Stains	72
	Z_Scratch	190
Missing	Bumps	0
	Dirtiness	0
	K_Scatch	0
	Other_Faults	0
	Pastry	0
	Stains	0
	Z_Scratch	0
Mean	Bumps	1523
	Dirtiness	1415
	K_Scatch	1384
	Other_Faults	1485
	Pastry	1554
	Stains	1358
	Z_Scratch	1358
Median	Bumps	1624
	Dirtiness	1364
	K_Scatch	1362
	Other_Faults	1372
	Pastry	1648
	Stains	1358
	Z_Scratch	1356
Standard deviation	Bumps	162
	Dirtiness	117
	K_Scatch	50.3
	Other_Faults	154
	Pastry	155
	Stains	1.49
	Z_Scratch	25.7
Minimum	Bumps	1227
	Dirtiness	1352
	K_Scatch	1336
	Other_Faults	1227
	Pastry	1333
	Stains	1354
	Z_Scratch	1348

Descriptives

Maximum	Bumps	1794
	Dirtiness	1696
	K_Scatch	1687
	Other_Faults	1715
	Pastry	1710
	Stains	1362
	Z_Scratch	1708

References

[1] The jamovi project (2025). *jamovi*. (Version 2.7) [Computer Software]. Retrieved from https://www.jamovi.org.

[2] R Core Team (2025). *R: A Language and environment for statistical computing*. (Version 4.5) [Computer software]. Retrieved from https://cran.r-project.org. (R packages retrieved from CRAN snapshot 2025-05-25).